

GraphMap: Scalable Crowd-Sourced Global Vectorized HD Map Construction Via Sparse Visual Graph Fusion

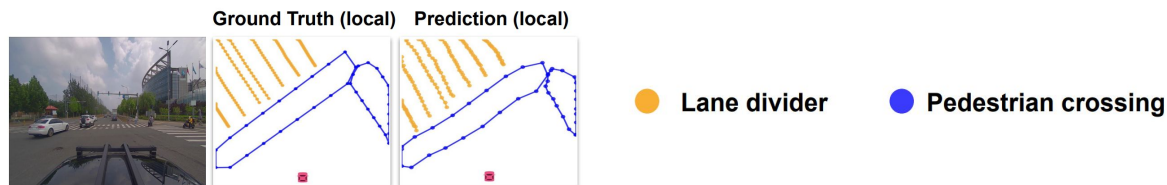
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Problem and Motivation

- Conventional HD map pipelines^[1,2] rely on collecting dense LiDAR point clouds and require *significant human annotation* (not scalable)
- Recent online HD mapping^[3, 4] use deep learning models to generate local vectorized HD maps around the ego-vehicle from onboard sensors
 - Works **only with short perception ranges** (e.g., 60×30 m)
 - Performance significantly drops with increasing range



Range	Method	Agent/Method	AP _{ped}	AP _{div}	mAP
30 × 30 m	MapTRv2	Vehicle-local	0.745	0.909	0.827
75 × 50 m	MapTRv2	Infra-local	0.772	0.785	0.778



Range	Method	Agent/Method	AP _{ped}	AP _{div}	mAP
120 × 60 m	MapTRv2	Vehicle-local	0.182	0.103	0.143
150 × 80 m	MapTRv2	Infra-local	0.399	0.482	0.440

[1] Lego-loam: Lightweight and ground-optimized lidar odometry and mapping on variable terrain, IROS 18

[2] Lio-sam: Tightly-coupled lidar inertial odometry via smoothing and mapping, IROS 20

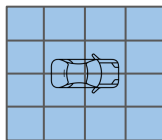
[3] MapTR: Structured Modeling and Learning for Online Vectorized HD Map Construction, ICLR 23

[4] Maptrv2: An end-to-end framework for online vectorized hd map construction, IJCV 24

Problem and Motivation

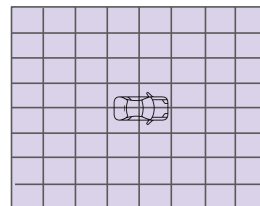
- Key question we try to answer
 - Can we **automatically** build **global vectoried maps** from multi-agent data?
- Why not using Collaborative Perception^[1, 2] methods?
 - Rely on ***dense raw data/feature map fusion*** → ***not scalable***
 - Computation/communication overhead is high for large feature maps
 - The feature map is centered at ***ego-vehicle and fixed-size***

30 m x 30 m → BEV feature (100 x 100)



3 km x 3 km → BEV feature (10000 x 10000)

Not feasible



[1] Robust collaborative 3d object detection in presence of pose errors, ICRA 23

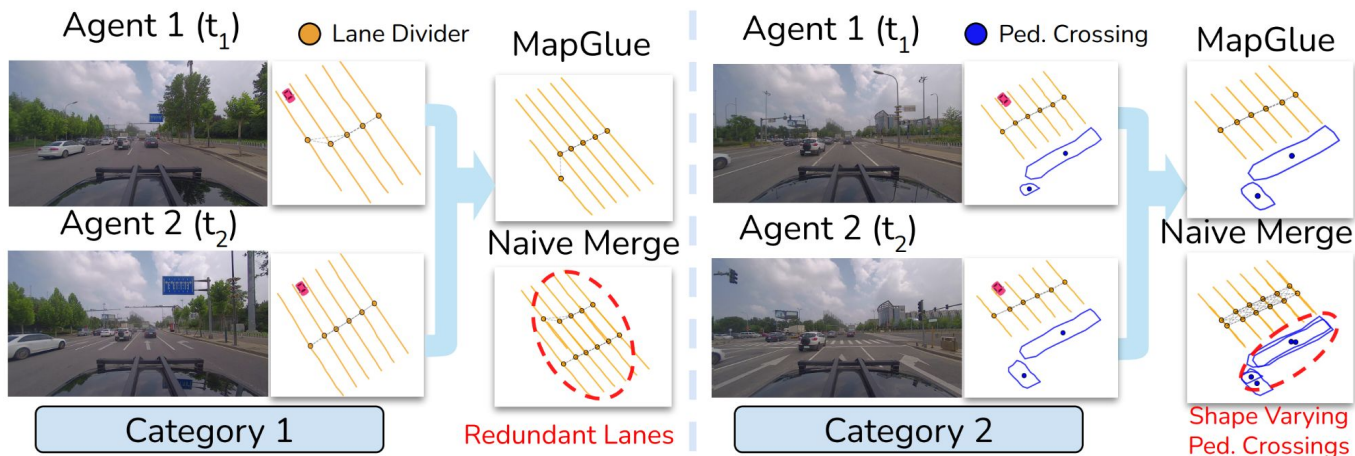
[2] Collaboration helps camera overtake lidar in 3d detection, CVPR 23

Solution Sketch

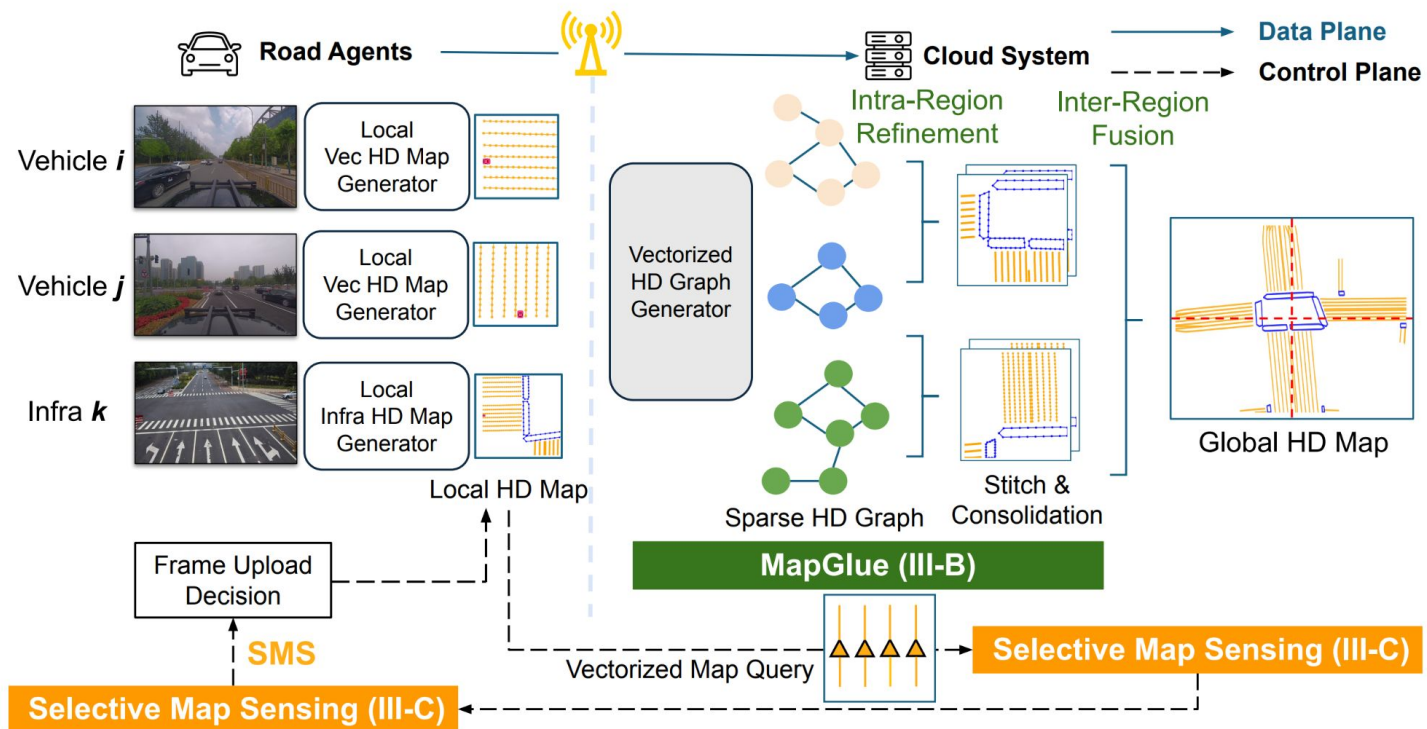
- Our approach (**GraphMap**)
 - Departing from dense, ego-centric fusion → **graph-based sparse fusion**
 - A vehicle-cloud crowd-sourced system leveraging to fuse *individual vectorized observations into global maps* accurately and efficiently
 - Vectorized observations are compact for transmitting over wireless network
 - Propose MapGlue: A Type-Aware Structure-Preserving Graph Matching algorithm
 - Reduce computation & bandwidth by Selective Map Sensing

Two Key Observations in Multi-Agent Map Fusion

- Category 1: Shape-Preserving with Spatial Offsets
 - *Lane Dividers, Road Boundaries, etc.*
- Category 2: Location-Preserving with Shape Variations
 - *Pedestrian Crossings, Junctions, etc.*



Design of GraphMap



Evaluation - End-to-end Map Accuracy

DAIR-V2X Maps: 06, 08, 09

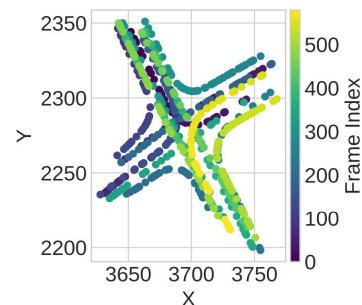
Map	Method	AP _{ped}	AP _{div}	mAP
<i>Range: 300 × 300 m</i>				
06	PolyMerge	0.148	0.085	0.117
	FDC	0.202	0.027	0.114
	HBM	0.167	0.094	0.130
	GraphMap (C)	0.875	0.303	0.589
	GraphMap (C+L)	0.875	0.320	0.598
<i>Range: 250 × 250 m</i>				
08	PolyMerge	0.080	0.137	0.074
	FDC	0.330	0.002	0.166
	HBM	0.000	0.229	0.114
	GraphMap (C)	1.000	0.502	0.751
	GraphMap (C+L)	1.000	0.481	0.741
<i>Range: 300 × 300 m</i>				
09	PolyMerge	0.769	0.058	0.413
	FDC	0.594	0.046	0.321
	HBM	0.070	0.050	0.060
	GraphMap (C)	0.923	0.077	0.500
	GraphMap (C+L)	0.923	0.166	0.545

DAIR-V2X Maps: 10, 13, 16

Map	Method	AP _{ped}	AP _{div}	mAP
<i>Range: 300 × 300 m</i>				
10	PolyMerge	0.000	0.183	0.091
	FDC	0.310	0.003	0.157
	HBM	0.000	0.119	0.060
	GraphMap (C)	0.857	0.416	0.636
	GraphMap (C+L)	0.857	0.457	0.647
<i>Range: 250 × 250 m</i>				
13	PolyMerge	0.618	0.081	0.399
	FDC	0.516	0.018	0.264
	HBM	0.575	0.063	0.319
	GraphMap (C)	1.000	0.088	0.544
	GraphMap (C+L)	1.000	0.163	0.581
<i>Range: 250 × 250 m</i>				
16	PolyMerge	0.494	0.017	0.256
	FDC	0.450	0.012	0.231
	HBM	0.125	0.098	0.111
	GraphMap (C)	0.833	0.099	0.466
	GraphMap (C+L)	0.833	0.140	0.528

C: Camera only
C+L: Camera+Lidar

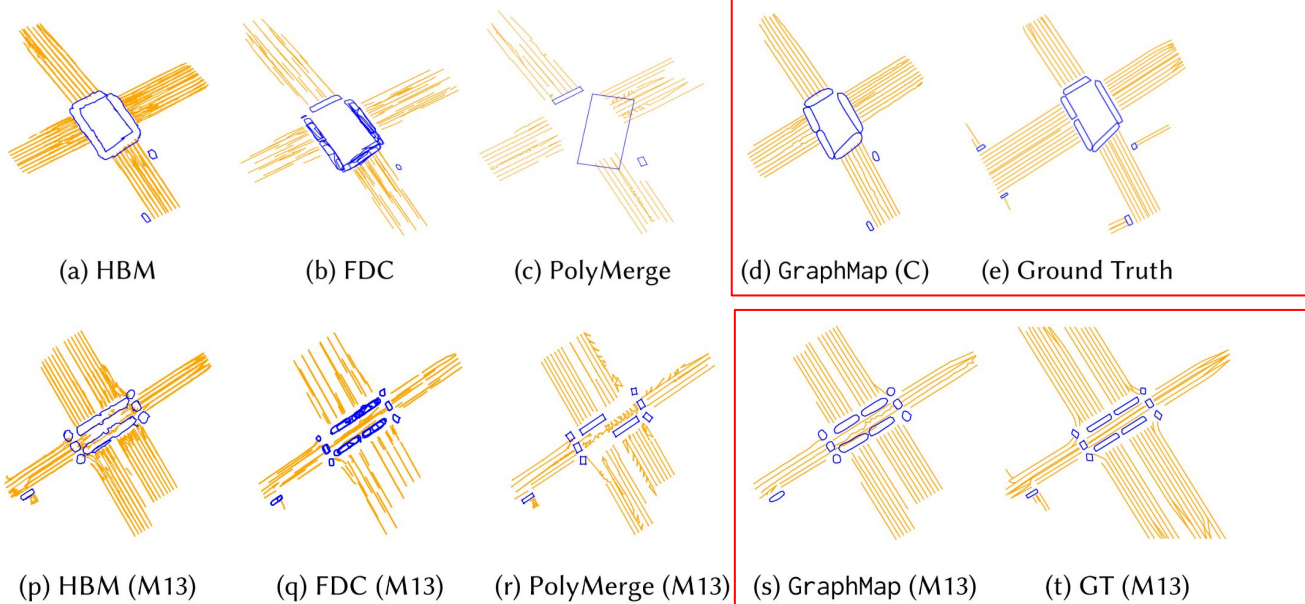
Agent Trajectories



(b) Map 09

Qualitative Visualization of the Global Maps

- GraphMap significantly outperforms baseline methods by producing a more fine-grained and accurate maps at larger scale



Thank You!



Our Team

